

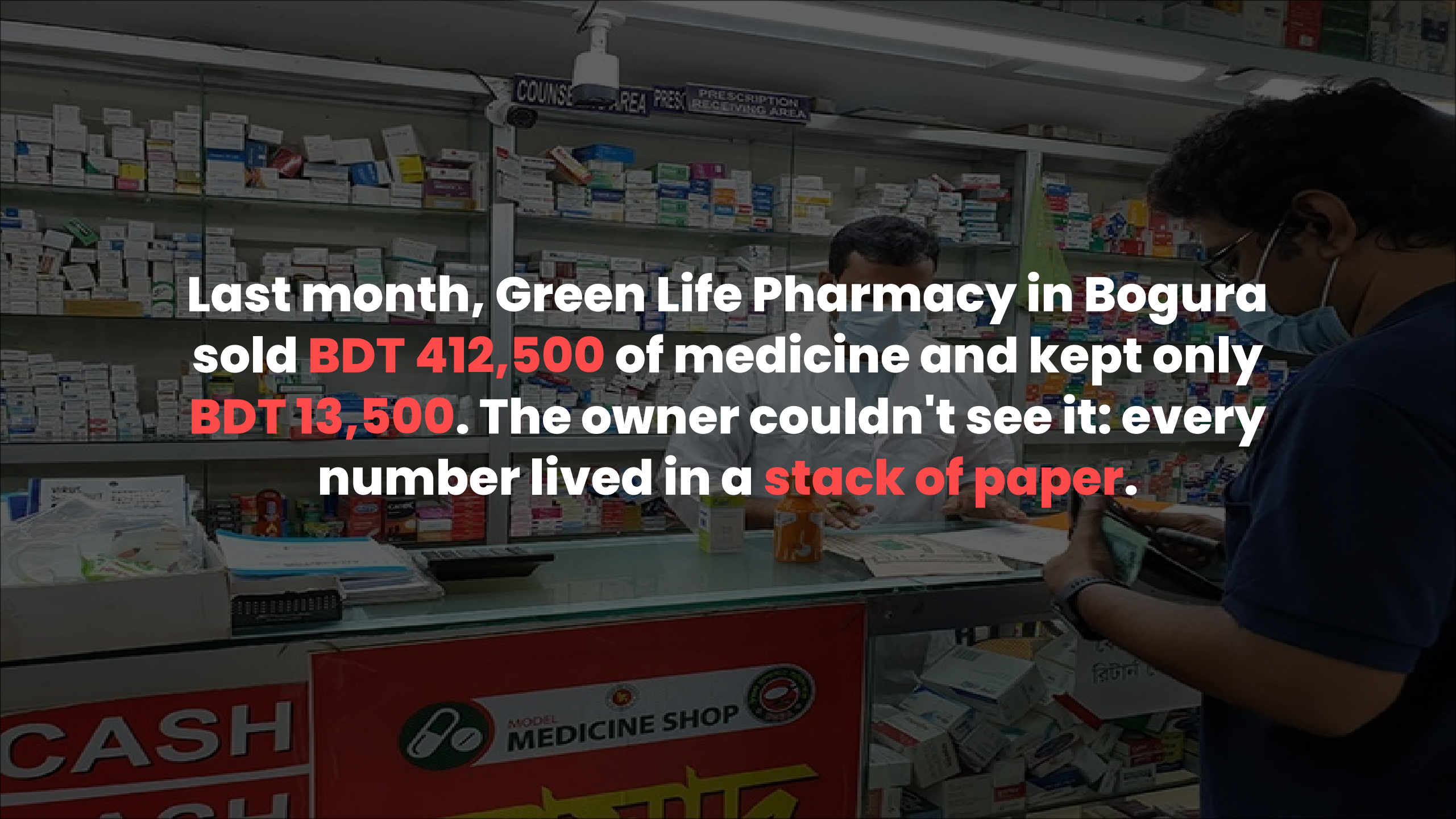
LaMP-Copilot

An AI finance officer for neighbourhood pharmacies in Bangladesh

Real accounts from paper records, early warning of cash trouble, and the cheapest way to fund every order.

FINTECHATHON 2026 · INTERNATIONAL TRACK · TOPIC E: SME FINANCE COPILOT

Team LaMP · Ishmam Tahmid (team lead) · Farhan Tahsin Khan



Last month, Green Life Pharmacy in Bogura sold BDT 412,500 of medicine and kept only BDT 13,500. The owner couldn't see it: every number lived in a stack of paper.

SITUATION ANALYSIS

The Manufacturing Triumph:

PHARMACEUTICAL MARKET

$ 3.3 Billion
market size, MAT Q1 2026

~11%
4-year growth (CAGR)

1.83%+ of GDP



98% of Demand Met Locally

About 213 working manufacturers make nearly all the medicine. Supply isn't the bottleneck; paying for it at the last mile is.

Credit flows informally and invisibly between makers and shops, at a cost no one measures.

The Last-Mile Retail Reality:



173,500+ Licensed Pharmacies

Plus up to 100,000 unlicensed shops, mostly small and owner-run.



Thin 10–15% Margins

Chemists are publicly demanding a higher commission from drug makers.



Forced Credit Dependency

Paper invoices and supplier credit from 10–50 drug companies each.



Expiry and Patient Credit

Unsold stock expires as pure loss; patients buy on credit too.

THE LAST MILE RUNS A MULTI-SUPPLIER CREDIT BUSINESS ON PAPER, WITH NO VIEW OF CASH, EXPIRY OR BORROWING COST

Sources: IQVIA MAT Q1 2026 (via industry reports); BAPI; DGDA (via Financial Express, Innovision); Springer; UNB.

MEET GREEN LIFE PHARMACY, BOGURA

The shop behind every number in this deck. Illustrative but consistent; our engines and tests reproduce each figure exactly.

Last 30 Days: Profit and Loss

Line	BDT	What it means
Sales	412,500	From the sales book
Gross profit at a 12% margin	49,500	After the cost of medicine
Rent, salaries, utilities	– 36,000	Fixed overheads
Net profit	**13,500**	What the owner keeps

Working Capital Snapshot

Item	Value	Read as
Stock on shelves	BDT 210,000	17.4 days of purchases
Owed by patients	BDT 38,000	2.8 days of sales
Active suppliers	23	23 sets of dues to track
Supplier invoices, 3 months	41	38 paid on time

BDT 18,400

of stock expires within 60 days

6 batches

BDT 9,200

can still go back to suppliers

68% of a month's profit

BDT 60,000

Supplier A stock-up order

due Mon 26 Oct

BDT 41,000

forecast cash on that day

short BDT 19,000

One expired batch or one unnecessary loan can wipe out a week of profit.

All of it lives in 41 paper invoices and a handwritten sales book, so not even the owner can see it.



PROBLEM IDENTIFICATION

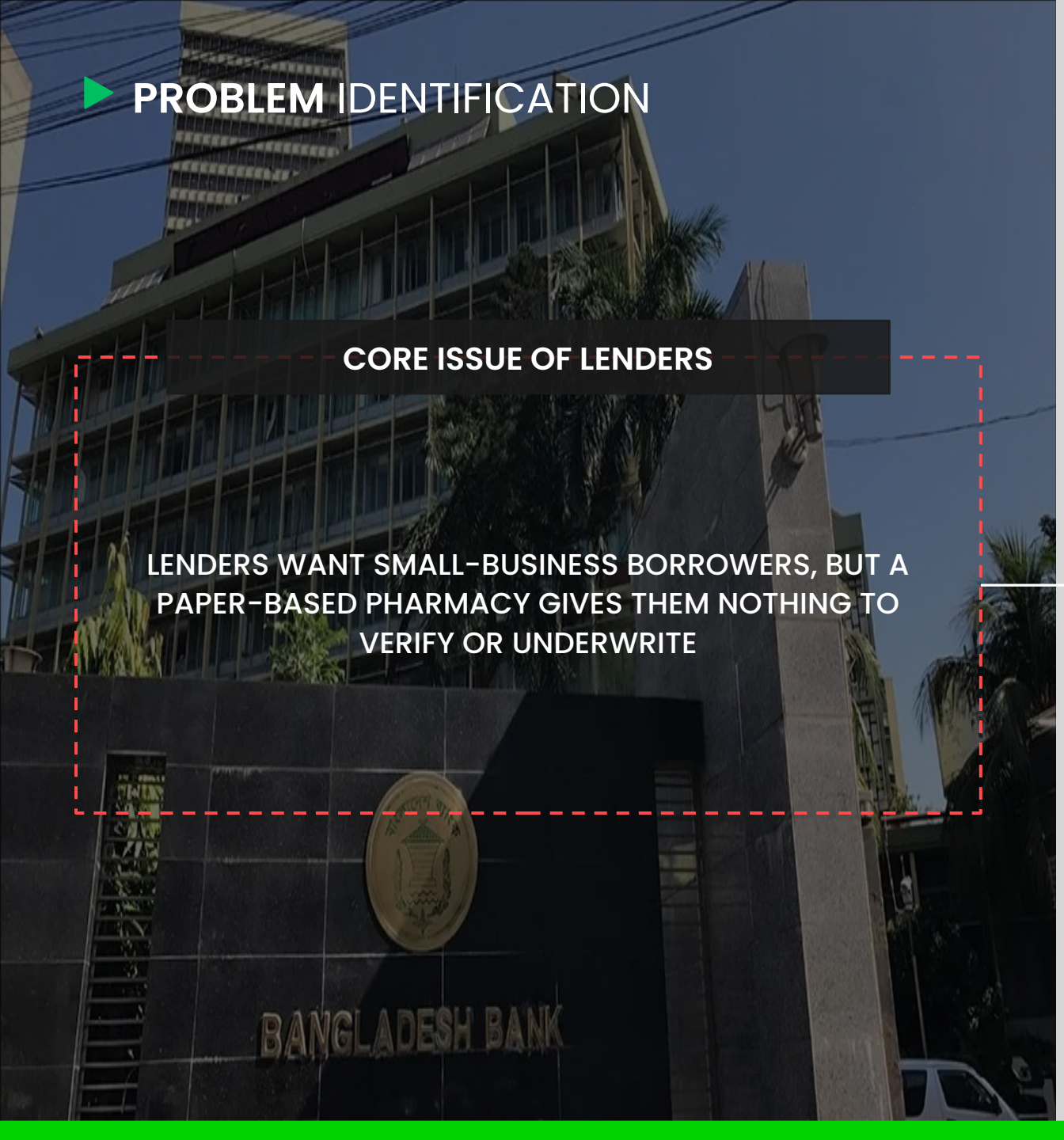
CORE ISSUE OF LOCAL PHARMACIES

PHARMACIES RUN A CREDIT BUSINESS WITH 10–50 SUPPLIERS ON PAPER, SO CASH CRUNCHES AND EXPIRY LOSSES STAY INVISIBLE UNTIL TOO LATE

(LOCAL PHARMACIES)

SYMPTOMS

- NO VIEW OF PROFIT, DUES OR STOCK: ONLY PAPER RECORDS
- CASH RUNS SHORT BEFORE BIG SUPPLIER PAYMENTS
- EXPIRED STOCK IS PURE LOSS; RETURNS GO UNCLAIMED
- PATIENTS BUY ON CREDIT WHILE THE SHOP OWES SUPPLIERS
- NO WAY TO TELL THE CHEAPEST LOAN, OR IF ONE IS NEEDED



PROBLEM IDENTIFICATION

CORE ISSUE OF LENDERS

LENDERS WANT SMALL-BUSINESS BORROWERS, BUT A PAPER-BASED PHARMACY GIVES THEM NOTHING TO VERIFY OR UNDERWRITE

(LENDERS)

SYMPTOMS

- NO AUDITED ACCOUNTS, TAX RETURNS OR DIGITAL RECORDS TO ASSESS
- CAN'T TELL IF A PAPER INVOICE IS REAL OR ALREADY FINANCED
- NO PRODUCT FOR A BDT 8,000–60,000 SUPPLIER GAP
- MUST LEND 27% TO CMSMES BY 2029, BUT THE SHARE FELL TO 15.9% IN MARCH 2026
- FINANCIAL DATA NEEDS EXPLICIT CONSENT UNDER THE PDPO 2025

A 3 WAY DEADLOCK



MANUFACTURERS

MUST SELL ON CREDIT
CASH TRAPPED IN RECEIVABLES



PHARMACIES

NEED MEDICINE EVERY DAY
CAN'T PAY UPFRONT OR SEE CASH AHEAD







LENDERS

WANT TO FINANCE
BUT CAN'T VERIFY RISK

Each side waits for another to move first: the pharmacy can't prove itself, the lender can't see it, and the manufacturer carries the gap.

CORE ISSUE STATEMENT

A pharmacy already produces the evidence of a working business: invoices, payments and expiry dates. But it is trapped on paper, so no one can act on it in time, not the owner and not a lender.

What's the actual solution?

LaMP-Copilot

An AI finance officer on the owner's phone: real accounts from paper, cash trouble spotted 30 days ahead, and every gap closed the cheapest way.

Justification

“The cheapest money is the cash you already have.” Trapped cash first, then free supplier time, and borrow only what's left.

OUR SOLUTION: ONE ANSWER FOR EACH PAIN

THE PAIN	WHAT LaMP DOES	RESULT AT GREEN LIFE
No financial picture Paper records; real profit unknown.	**Reads invoice and sales-book photos into real accounts** The owner confirms every figure.	**BDT 13,500** monthly profit, finally visible
Expiry losses Returns go unclaimed.	**Tracks every batch's expiry and return window** Drafts return claims with the invoice as evidence.	**BDT 9,200** recovered: 68% of monthly profit
Cash crunch before big orders The order is due before the cash.	**Forecasts 30 days across all 23 suppliers** Trims slow movers; asks for more days.	**19,000 → 8,200** shortfall, spotted 10 days early
No fair finance Lenders can't verify paper.	**Prices every option in taka; builds a credit passport** Shared only with chosen lenders.	**BDT 0–182** instead of BDT 422

Every taka comes from deterministic code. The AI never computes or moves money.

How LaMP Works (5 Sequential Steps)

1. Capture (Turn Paper into Data)

Owner's Phone · Pharmacy Counter

The owner photographs invoices and sales-book pages; an OpenAI vision model reads batches, expiry dates, quantities, prices and due dates.

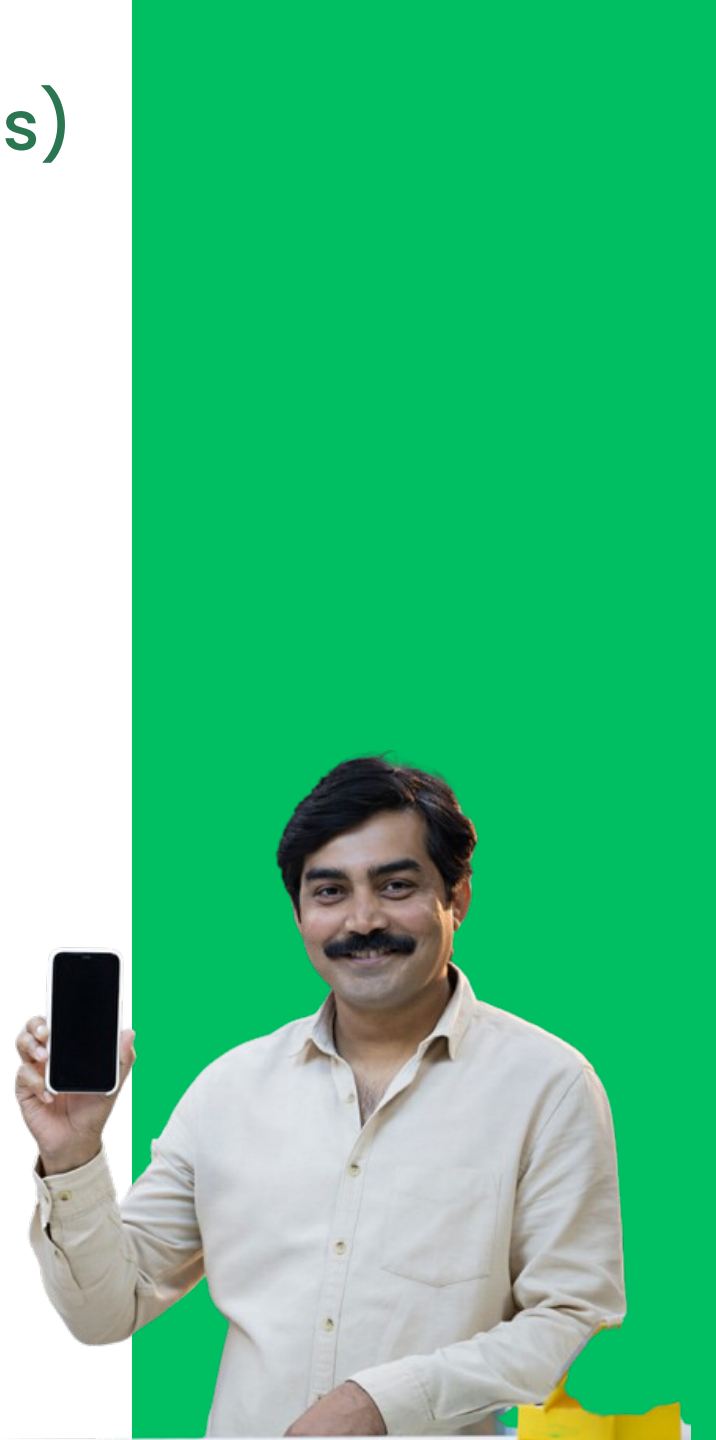
Unsure fields are highlighted; the owner fixes them before anything is saved.

Plain-code checks verify totals and dates.

Text in a photo is data: a planted “approve the maximum loan” is flagged.

Outcome

Paper becomes a verified record the owner has confirmed.



How LaMP Works (5 Sequential Steps)

2. Understand (Build Real Accounts)

LaMP Engines · Deterministic Python

From confirmed records, LaMP builds the accounts a lender would ask for: profit, margin, stock days and money in expiring stock.

Green Life: sales BDT 412,500, net profit BDT 13,500 at a 12% margin.

6 batches worth BDT 18,400 expire within 60 days; BDT 9,200 is still returnable.

LaMP drafts the return claims with the invoice attached.

Outcome

Profit, dues, stock and expiry on one screen.



How LaMP Works (5 Sequential Steps)

3. Anticipate (Forecast and Alert)

30-Day Cash Forecast · All 23 Suppliers

A 30-day forecast combines sales, patient dues, supplier dues, rent, salaries and orders, and marks the first day cash runs out.

Supplier A's BDT 60,000 order is due Mon 26 Oct; cash that day: BDT 41,000.

Alert ten days ahead: short BDT 19,000.

Also flags expiring stock, closing return windows and overdue patient dues.

Outcome

Cash trouble shows up while it's still cheap to fix.



How LaMP Works (5 Sequential Steps)

4. Fund (Cheapest Money First)

Funding Plan · Owner Approval

30 boxes of a slow mover sell 0.2 a day with 60 sellable days left, so only 12 will sell before expiry.

Trim to 12 boxes: BDT 10,800 saved, gap down from BDT 19,000 to BDT 8,200.

Options in taka: supplier extension free, wallet loan BDT 182 (2.2% of the gap), supplier credit BDT 492 (6.0%).

Plan: ask for 20 more days; no reply by Fri 23 Oct and LaMP proposes the loan. The owner approves with a PIN.

Outcome

BDT 0–182 instead of BDT 422, and no stock left to expire.



How LaMP Works (5 Sequential Steps)

5. Prove (Credit Passport and Consent)

Lender View · Sandbox Partners

38 of 41 supplier-confirmed invoices paid on time become a credit passport: on-time estimate 0.826, default risk 3%.

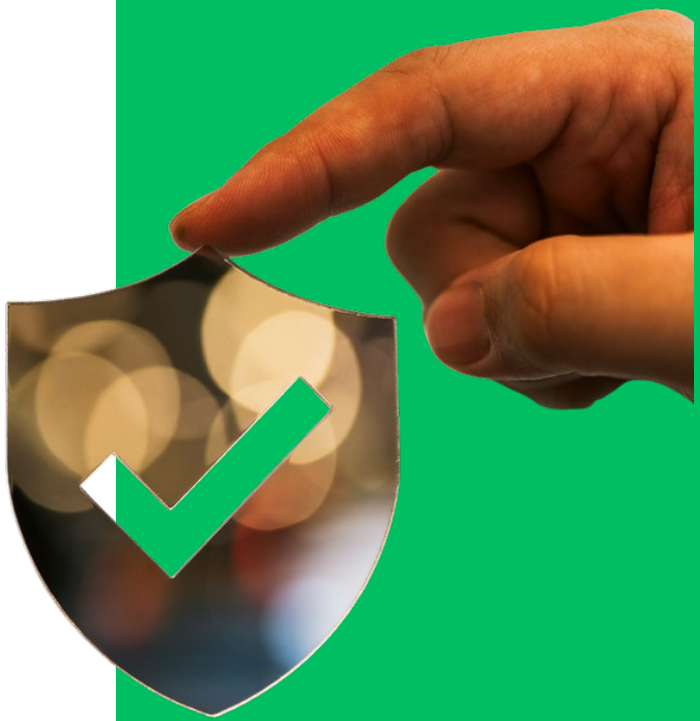
Shelf-life credit: a purpose-blind lender funds BDT 19,000; LaMP funds BDT 8,200, the stock that sells.

A 75.2% advance offer; a 35% repayment ceiling refuses loans the shop can't carry.

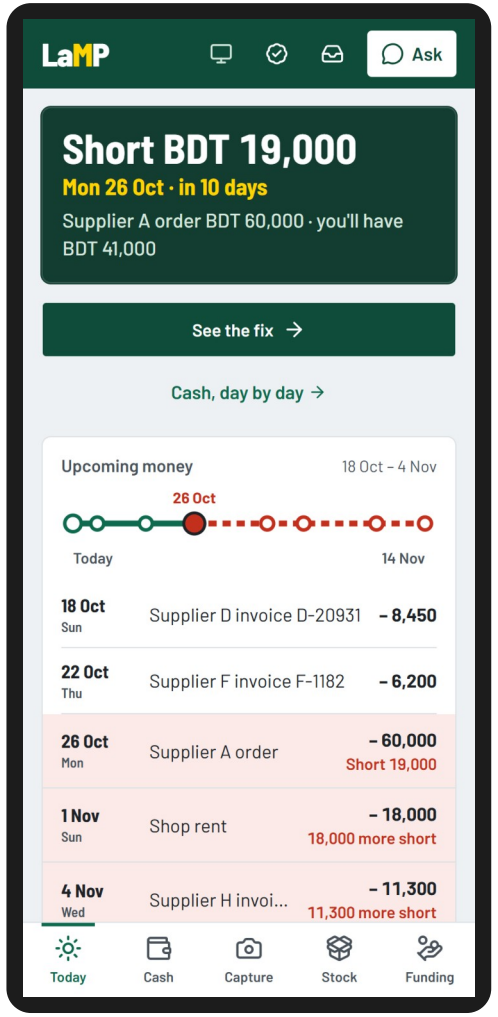
Shared per lender and scope, withdrawable (PDPO 2025); a registry stops double financing.

Outcome

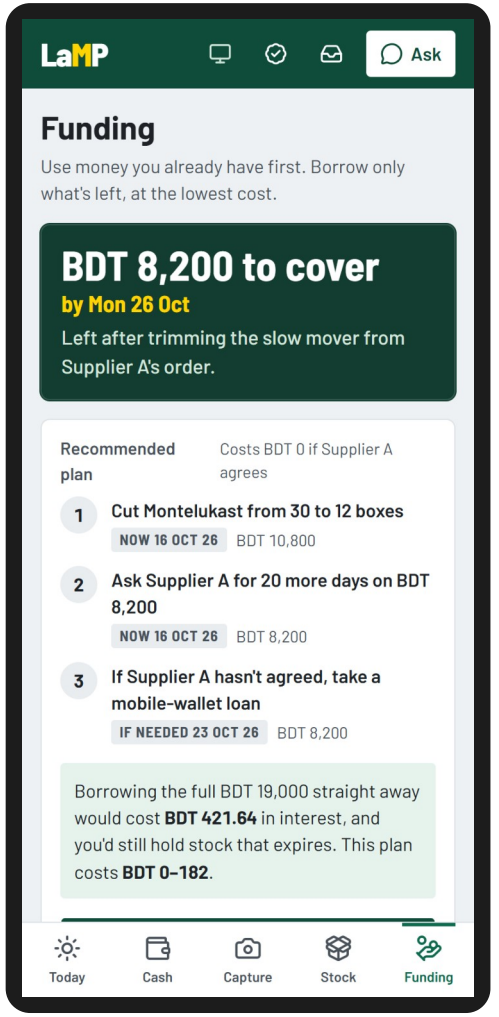
A verifiable, consented path to fair finance.



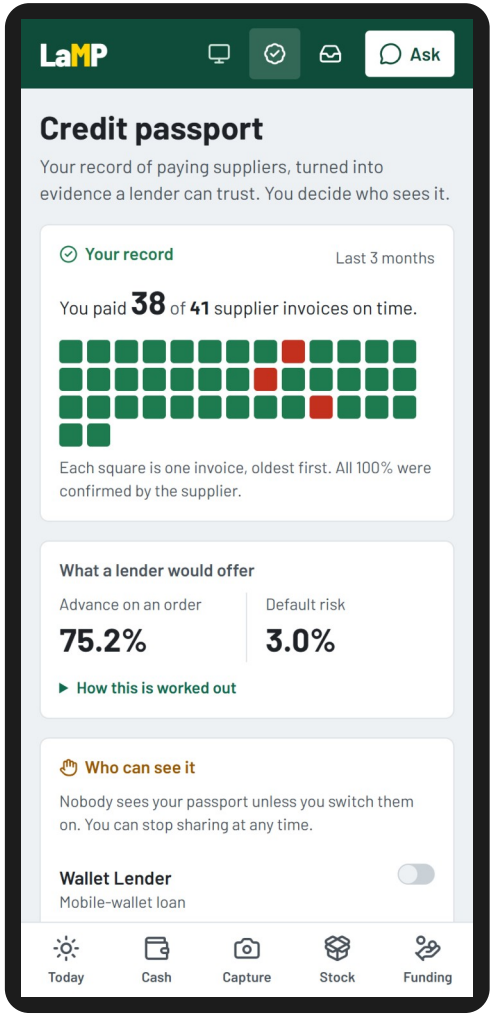
THE COPILOT, LIVE ON GREEN LIFE'S DATA



Today



Funding



Credit passport

What the owner sees

Today: The banner names the next shortfall: Short BDT 19,000, Mon 26 Oct, in 10 days.

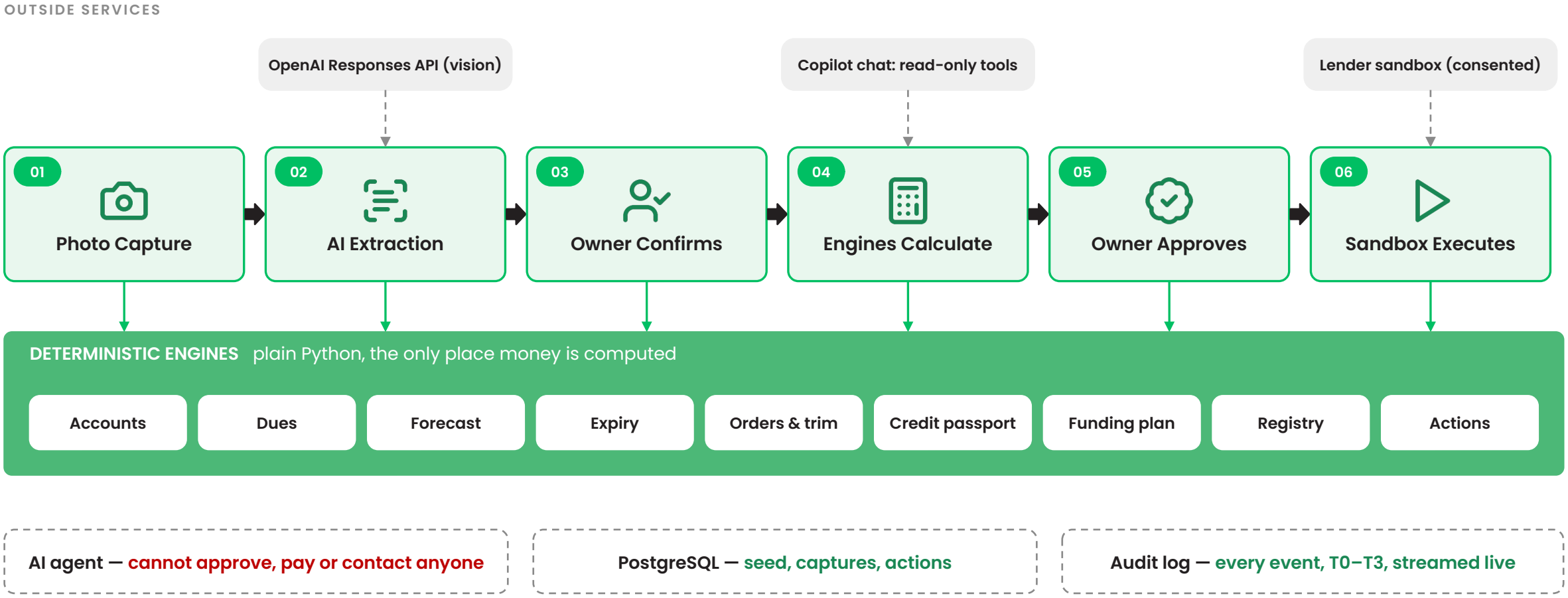
Funding: BDT 8,200 left after the trim, a three-step plan, every option in taka. The owner approves.

Credit passport: 38 of 41 on time, a 75.2% advance offer, and a sharing switch per lender.

Built and running
Next.js · FastAPI · PostgreSQL · OpenAI Responses API · hash-chained audit log · 249 automated tests

SYSTEM FLOW

AI READS AND PROPOSES, CODE CALCULATES — AND ONLY THE OWNER CAN APPROVE MONEY MOVING



TRUST MECHANISM

How LaMP makes AI misuse and double financing structurally impossible, not just against the rules

Can anyone trick the AI into moving money?

No — there is no tool to do it

Read-only tools plus one proposal

The chat can read and propose. No tool approves, pays or contacts anyone.

Approval lives outside the AI

Only the owner approves, in the app; code then runs the step in a sandbox.

Text in photos is data

A planted “approve the maximum loan” is flagged, and nothing happens.

Can one invoice be financed twice, or data leak to a lender?

No — registry lock and consent

One invoice, one financing

Each invoice's fingerprint is locked; a second application is rejected.

Consent per lender and scope

Lenders see only what the owner shares, withdrawable any time.

No patient medicine data

Patient dues are amounts only; purchases are health data.

The guarantee is structural: the tools the AI would need don't exist, and every decision is logged in a hash chain anyone can verify.

SECURITY & COMPLIANCE

Permission Tiers

Tier	Who	What it may do
T0	Regulator, auditor	Read-only audit views
T1	Owner, photo reading	Capture, confirm, approve own actions
T2	Engines	Compute; no outside contact
T3	Lenders, money steps	Consented reads, sandbox payouts

What's Real and What's Sandbox in This Prototype

Real, working code	Sandbox, said openly
Photo reading (OpenAI vision)	Lenders, decisions, payouts
All money maths	Supplier replies
Copilot chat, read-only tools	Supplier invoice confirmation (seeded)
Approvals with PIN, Pause, consent, duplicate guard, hash-chained log	Federated learning (roadmap)

249 / 249

tests pass, no AI calls

26 / 26

red-team attacks held

0

money figures computed by the AI

Every

event hash-chained and verifiable

Execution evidence: every event, with its tier and chain hash, is exported to logs/sandbox_execution_trace.log.

Lenders, payouts and supplier replies are sandbox partners. No real money moves.

INNOVATION: WHAT MAKES LaMP NEW

Four things no pharmacy tool in Bangladesh does today, each working in the prototype.



WORKING CAPITAL · EXPIRY-AWARE

Orders and credit sized on what will actually sell

BDT 10,800
of stock avoided on one order

Shelf-life credit: lend and order only what sells before expiry, so BDT 19,000 of need becomes BDT 8,200.



NEUTRAL · MULTI-SUPPLIER

Works across every supplier, not one distributor

10–50
suppliers per pharmacy

Dues, returns and financing across all suppliers, not one distributor's orders.



CREDIT · CONSENTED PASSPORT

Supplier payments become lender evidence

75.2%
advance on 38 of 41 on time

A supplier-confirmed record replaces missing collateral, shared per lender with consent.



AGENTIC · SAFE BY DESIGN

An agent that acts, without touching money

0
money figures computed by the AI

It reads, forecasts and prepares actions; code computes every taka, and the owner approves.

No one else combines all four: pharmacy-specific, neutral across suppliers, consented, and safe by design.

COMPETITIVE LANDSCAPE

Others prove the demand; none is neutral, pharmacy-aware, agentic and capital-light at once.

Player	Model	Neutral across suppliers	Knows batches and expiry	Acts, not just records	No stock or capital on its books
PulseTech (Bangladesh)	Distributor finance; 12,000+ Dhaka pharmacies	**No: own orders**	**Yes**	**Partly**	**No**
TallyKhata (Bangladesh)	Bookkeeping for any merchant; 4.4M users	**Yes**	**No: any shop**	**No**	**Yes**
Shelf Life (Nigeria, Kenya)	Pay-as-you-sell pharmacy stock	**No**	**Yes**	**Partly**	**No: holds stock**
Retailio (India)	Marketplace credit; 100,000+ pharmacies	**No: own marketplace**	**Partly**	**Partly**	**Yes**
LaMP-Copilot	AI finance officer across all suppliers	**Yes**	**Yes**	**Yes**	**Yes**

vs PulseTech
Neutral across all 10–50 suppliers.

vs TallyKhata
Pharmacy-specific and agentic.

vs Shelf Life
No stock or capital on our books.

Sources: Financial Express, Startup Bangladesh, The Daily Star, Accion, TechCabal, BioSpectrum, FinBox.

REVENUE MODEL

Primary: an origination fee from partner lenders. **Second:** manufacturer and distributor programmes.

Who Pays LaMP

Pharmacy

Pays nothing; free is our adoption lever.

Partner lender

Gets verified, consented borrowers.

Manufacturer or distributor

Gets faster collections and evidence-backed returns.

Revenue Streams (proposed, to be validated in the pilot)

Stream	Payer	Charged on	Proposed rate
Origination fee	Partner lender	Amount advanced	**1.0%**
Programme subscription	Manufacturer or distributor	Per company, per month	**BDT 2.5–3.5 lakh**
Pharmacy app	Pharmacy	—	**Free**

Pricing Logic

Lenders already pay to acquire borrowers; a verified applicant is cheaper to approve.

Recovered expiry and on-time collections are worth more than the subscription.

Why this model is capital-light

LaMP never lends or holds funds. Lenders supply capital and carry risk; we supply verification. No stock, no loan book.

FEASIBILITY

	RISK EVENTS	MITIGATION	CONTINGENCY	RISK LEVEL
A	Owners skip invoices	Capture takes a minute and pays off at once	Distributors upload e-invoices	**Medium**
B	AI misreads a figure	Owner confirms every field; code checks totals	Manual entry fallback	**Medium**
C	Lenders distrust the record	Supplier-confirmed invoices, cautious score	Shadow mode with one lender first	**High**
D	Privacy breach or AI misuse	Consent per lender, no patient data, no money tools	Legal review; all 26 attacks in our red-team run held	**High**

Action Plan & Roadmap

From one pharmacy in Bogura to nationwide impact

Prove it with real invoices and repayments, grow across suppliers, then become the neutral finance layer.

1 Pilot (0–6 months)

Prove the model with 20 pharmacies in Bogura.

Pharmacies → Copilot → Lender

Key Actions

- 20 pharmacies, one distributor
- One lender, shadow mode first
- Measure accuracy and forecast error

Success Metrics (targets)

- 80%+ of invoices captured
- First loans repaid on time

2 Network (6–18 months)

Grow across suppliers and districts.

Distributors → More shops → Passports

Key Actions

- Distributor programmes
- Bangla interface
- Passports with 3+ lenders

Success Metrics (targets)

- 500+ pharmacies
- 3+ distributors

3 Scale (18–36 months)

Become the neutral finance layer for pharmacies.

Banks → Shared risk → Nationwide

Key Actions

- Bank supplier-credit product
- Federated risk model across lenders
- Nationwide rollout

Success Metrics (targets)

- 5,000+ pharmacies
- BDT 100M+ financed

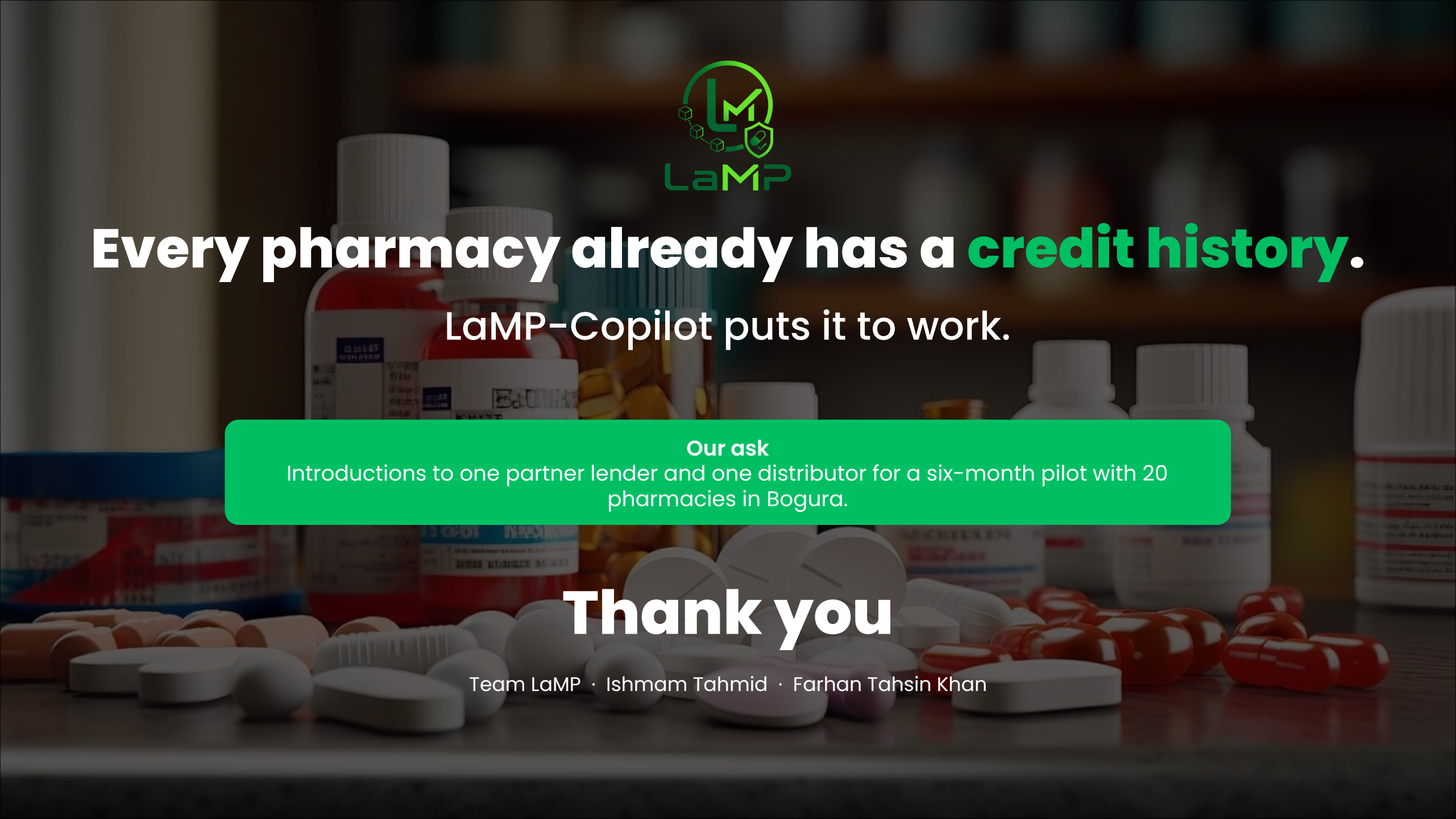
Long-Term Impact

Bankable pharmacies
Verified records replace missing collateral

Less expired medicine
Orders sized on what will sell

Cheaper finance
Own cash first, cheapest credit last

Healthier Bangladesh
Fewer stock-outs, more medicine on shelves



LaMP

Every pharmacy already has a credit history.

LaMP-Copilot puts it to work.

Our ask
Introductions to one partner lender and one distributor for a six-month pilot with 20 pharmacies in Bogura.

Thank you

Team LaMP · Ishmam Tahmid · Farhan Tahsin Khan